BITS Pilani WILP — Deep Reinforcement Learning

Assignment 1 - Part 1: Multi-Armed Bandit

Group : 87

Problem : Adaptive Treatment Recommendation System

Filename : Team 87 — MAB.pdf

Team Members

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Section 1 — Imports, Seeds & Group Configuration

```
# Import std libraries
import random
import numpy as np
import pandas as pd

import matplotlib.pyplot as plt
import matplotlib.patches as mpatches

import warnings
warnings.filterwarnings('ignore')

%matplotlib inline

# Group number
G = 87

# Set random seeds for repeatability
random.seed(G)
np.random.seed(G)

print("Group Number : ",G)
print(f"Seeds set    : random.seed({G}), numpy.random.seed({G})")
```

```
Group Number :  87
Seeds set    : random.seed(87), numpy.random.seed(87)
```

Task 1 — Dataset Design

Generate synthetic patient-treatment dataset for Group 87. Formula derivations:

- K = (G mod 3) + 5
- P_i = 0.4 + ((G + i) mod 6) x 0.07
- Severity = (patient_id mod 5) + 1
- UtilityScore = clinical_outcome x (1 – Severity/10)

```
def compute_parameters(G):
    # Calculate # of medicines and their hidden success probabilities with K=(G mod 3)+5
    K = (G % 3) + 5
    probs = [0.4 + ((G + i) % 6) * 0.07 for i in range(K)] # Hidden Prob
    return K, probs

def generate_patient_dataset(n=1000):
    # Create 1000 patients with a severity score cycling from 1 to 5
    ids = list(range(n))
    severities = [(pid % 5) + 1 for pid in ids]
    return pd.DataFrame({"patient_id": ids, "severity_score": severities})

def simulate_treatment(patient_row, medicine, hidden_probs):
    p               = hidden_probs[medicine]
    outcome         = int(np.random.rand() < p)       # 1 = recovered, 0 = not
    severity        = patient_row["severity_score"]
```

```
    utility         = outcome * (1 - severity / 10.0) # reward decreases with severity
    return outcome, utility

# Compute group parameters
K, hidden_probs = compute_parameters(G)

print(f"Group G = {G}")
print(f"Number of Medicines K = {K}")
print()
print("Hidden Success Probabilities:")
for i, p in enumerate(hidden_probs):
    tag = "  <-- OPTIMAL" if p == max(hidden_probs) else ""
    print(f"  Medicine {i}: P = {p:.4f}{tag}")

print(f"\nTrue Best Medicine: {int(np.argmax(hidden_probs))} (P = {max(hidden_probs):.4f})")

# Generate the patient dataset
base_df = generate_patient_dataset()
print("\nFirst 10 Patient Records:")
print(base_df.head(10).to_string(index=False))
print(f"\nTotal Patients : {len(base_df)}")
print(f"Severity Counts: {base_df['severity_score'].value_counts().sort_index().to_dict()}")
```

```
Group G = 87
Number of Medicines K = 5

Hidden Success Probabilities:
  Medicine 0: P = 0.6100
  Medicine 1: P = 0.6800
  Medicine 2: P = 0.7500  <-- OPTIMAL
  Medicine 3: P = 0.4000
  Medicine 4: P = 0.4700

True Best Medicine: 2 (P = 0.7500)

First 10 Patient Records:
 patient_id  severity_score
          0               1
          1               2
          2               3
          3               4
          4               5
          5               1
          6               2
          7               3
          8               4
          9               5

Total Patients : 1000
Severity Counts: {1: 200, 2: 200, 3: 200, 4: 200, 5: 200}
```

Task 2 — Immediate Exploitation Strategy

Policy: Test each medicine 10 times -> exploit the one with highest observed success rate.

```
# Immediate Exploitation Strategy
def greedy_exploitation(base_df, hidden_probs, K, warm_up=10):
    df = base_df.copy()
    df["assigned_medicine"] = -1
    df["clinical_outcome"]  = 0
    df["utility_score"]     = 0.0

    counts    = [0] * K   # how many times each medicine was tried
    successes = [0] * K   # how many successful recoveries per medicine

    total_reward       = 0.0
    cumulative_rewards = []

    for i in range(len(df)):
        row = df.iloc[i]

        if i < K * warm_up:
            # Warm-up phase: cycle through all medicines equally
            medicine = i % K
        else:
            # Exploit phase: pick the medicine with the best success rate
            rates    = [successes[m] / counts[m] if counts[m] > 0 else 0.0 for m in range(K)]
            medicine = int(np.argmax(rates))

        outcome, utility = simulate_treatment(row, medicine, hidden_probs)
```

```
            counts[medicine]    += 1
            successes[medicine] += outcome

            df.at[i, "assigned_medicine"] = medicine
            df.at[i, "clinical_outcome"]  = outcome
            df.at[i, "utility_score"]     = utility

            total_reward += utility
            cumulative_rewards.append(total_reward)

        # Print results
        rates = [successes[m] / counts[m] if counts[m] > 0 else 0.0 for m in range(K)]
        best  = int(np.argmax(rates))
        print("Task 2 - Greedy Strategy Results")
        print(f"  Final Cumulative Reward  : {total_reward:.4f}")
        print(f"  Times Each Medicine Used : {counts}")
        print(f"  Observed Success Rates   : {[round(r, 3) for r in rates]}")
        print(f"  Best Medicine Found      : {best}")
        print(f"  True Best Medicine       : {int(np.argmax(hidden_probs))}")

        return df, cumulative_rewards

    np.random.seed(G)
    greedy_df, greedy_rewards = greedy_exploitation(base_df, hidden_probs, K)

    # Show a few rows of the result
    print("\nSample Output (first 5 patients + first exploit patient):")
    sample = pd.concat([greedy_df.head(5), greedy_df.iloc[[K * 10]]])
    print(sample[["patient_id", "severity_score", "assigned_medicine",
                  "clinical_outcome", "utility_score"]].to_string(index=False))
```

```
Task 2 - Greedy Strategy Results
  Final Cumulative Reward  : 518.9000
  Times Each Medicine Used : [10, 16, 954, 10, 10]
  Observed Success Rates   : [0.3, 0.625, 0.753, 0.5, 0.5]
  Best Medicine Found      : 2
  True Best Medicine       : 2

Sample Output (first 5 patients + first exploit patient):
 patient_id  severity_score  assigned_medicine  clinical_outcome  utility_score
          0               1                  0                 0            0.0
          1               2                  1                 0            0.0
          2               3                  2                 0            0.0
          3               4                  3                 1            0.6
          4               5                  4                 1            0.5
         50               1                  1                 0            0.0
```

Task 3 — Controlled Clinical Trial Strategy

Policy: With probability ε explore (random medicine), with probability 1-ε exploit (best so far).

Simulates a clinical setting where most patients get the current best treatment, but a controlled fraction are enrolled in exploratory trials. Tested at ε ∈ {1%, 10%, 50%}.

```
    # Epsilon-Greedy Strategy

    def epsilon_greedy(base_df, hidden_probs, K, epsilon=0.10):
        df = base_df.copy()
        df["assigned_medicine"] = -1
        df["clinical_outcome"]  = 0
        df["utility_score"]     = 0.0

        counts    = [0] * K
        successes = [0] * K

        total_reward       = 0.0
        cumulative_rewards = []

        for i in range(len(df)):
            row = df.iloc[i]

            # Explore: pick a random medicine
            # Exploit: pick the medicine with the best success rate so far
            if np.random.rand() < epsilon or all(c == 0 for c in counts):
                medicine = np.random.randint(0, K)
            else:
                rates    = [successes[m] / counts[m] if counts[m] > 0 else 0.0 for m in range(K)]
                medicine = int(np.argmax(rates))

            outcome, utility = simulate_treatment(row, medicine, hidden_probs)
```

```
        counts[medicine]    += 1
        successes[medicine] += outcome

        df.at[i, "assigned_medicine"] = medicine
        df.at[i, "clinical_outcome"]  = outcome
        df.at[i, "utility_score"]     = utility

        total_reward += utility
        cumulative_rewards.append(total_reward)

    return df, cumulative_rewards, counts, successes

# Run for all three epsilon values
epsilon_configs = [(0.10, "10%"), (0.01, "1%"), (0.50, "50%")]
eps_results     = {}

print("Task 3 - Epsilon-Greedy Results")
print(f"  {'Epsilon':>8}  {'Final Reward':>14}  {'Medicine Counts':>30}  {'Best Found':>10}")
print("  " + "-" * 70)

for eps, label in epsilon_configs:
    np.random.seed(G)   # reset seed so each run is comparable
    df_e, rew_e, cnt_e, suc_e = epsilon_greedy(base_df, hidden_probs, K, epsilon=eps)
    eps_results[label]  = {"df": df_e, "rewards": rew_e, "counts": cnt_e, "successes": suc_e}
    best = int(np.argmax([suc_e[m] / cnt_e[m] if cnt_e[m] > 0 else 0 for m in range(K)]))
    print(f"  {label:>8}  {rew_e[-1]:>14.4f}  {str(cnt_e):>30}  {best:>10}")

print()
print("Analysis:")
print("  epsilon=1%  : Almost always exploits. Locks in fast but may pick the wrong medicine.")
print("  epsilon=10% : Good balance. Keeps exploring a little while using the best medicine.")
print("  epsilon=50% : Explores too much. Ignores what it already learned. Very wasteful.")
```

```
Task 3 - Epsilon-Greedy Results
   Epsilon    Final Reward                Medicine Counts  Best Found
  ----------------------------------------------------------------------
       10%        505.9000          [30, 216, 699, 31, 24]           2
        1%        394.9000             [840, 6, 0, 150, 4]           0
       50%        460.1000         [92, 134, 580, 108, 86]           2

Analysis:
  epsilon=1%  : Almost always exploits. Locks in fast but may pick the wrong medicine.
  epsilon=10% : Good balance. Keeps exploring a little while using the best medicine.
  epsilon=50% : Explores too much. Ignores what it already learned. Very wasteful.
```

Task 4 — Confidence-Based Strategy - UCB1

```
# UCB1 Strategy
def ucb1_strategy(base_df, hidden_probs, K):
    df = base_df.copy()
    df["assigned_medicine"] = -1
    df["clinical_outcome"]  = 0
    df["utility_score"]     = 0.0

    counts      = [0]   * K   # number of times each medicine was tried
    total_util  = [0.0] * K   # total utility collected per medicine
    mean_util   = [0.0] * K   # average utility per medicine

    total_reward       = 0.0
    cumulative_rewards = []

    for t in range(1, len(df) + 1):
        row = df.iloc[t - 1]

        if t <= K:
            # Try each medicine once first to avoid dividing by zero
            medicine = t - 1
        else:
            # Calculate UCB score for each medicine and pick the highest
            ucb_scores = [
                mean_util[m] + np.sqrt(2 * np.log(t) / counts[m])
                if counts[m] > 0 else float("inf")
                for m in range(K)
            ]
            medicine = int(np.argmax(ucb_scores))

        outcome, utility = simulate_treatment(row, medicine, hidden_probs)

        counts[medicine]     += 1
```

```
            total_util[medicine] += utility
            mean_util[medicine]   = total_util[medicine] / counts[medicine]

            df.at[t - 1, "assigned_medicine"] = medicine
            df.at[t - 1, "clinical_outcome"]  = outcome
            df.at[t - 1, "utility_score"]     = utility

            total_reward += utility
            cumulative_rewards.append(total_reward)

        best = int(np.argmax(mean_util))
        print("Task 4 - UCB1 Strategy Results")
        print(f"  Final Cumulative Reward  : {total_reward:.4f}")
        print(f"  Times Each Medicine Used : {counts}")
        print(f"  Mean Utility per Medicine: {[round(u, 4) for u in mean_util]}")
        print(f"  Best Medicine Found      : {best}")
        print(f"  True Best Medicine       : {int(np.argmax(hidden_probs))}")

        return df, cumulative_rewards, counts, mean_util

    np.random.seed(G)
    ucb_df, ucb_rewards, ucb_counts, ucb_means = ucb1_strategy(base_df, hidden_probs, K)

    # Show how often each medicine was selected
    print("\nMedicine Selection Counts (UCB1):")
    for m in range(K):
        bar = "#" * (ucb_counts[m] // 10)
        print(f"  Medicine {m} (P={hidden_probs[m]:.2f}): {ucb_counts[m]:>4} times  {bar}")
```

```
Task 4 - UCB1 Strategy Results
  Final Cumulative Reward  : 449.1000
  Times Each Medicine Used : [183, 233, 400, 99, 85]
  Mean Utility per Medicine: [np.float64(0.4273), np.float64(0.4592), np.float64(0.5157), np.float64(0.3263), np.float64(0.2
  Best Medicine Found      : 2
  True Best Medicine       : 2

Medicine Selection Counts (UCB1):
  Medicine 0 (P=0.61):  183 times  ##################
  Medicine 1 (P=0.68):  233 times  #######################
  Medicine 2 (P=0.75):  400 times  ########################################
  Medicine 3 (P=0.40):   99 times  #########
  Medicine 4 (P=0.47):   85 times  ########
```

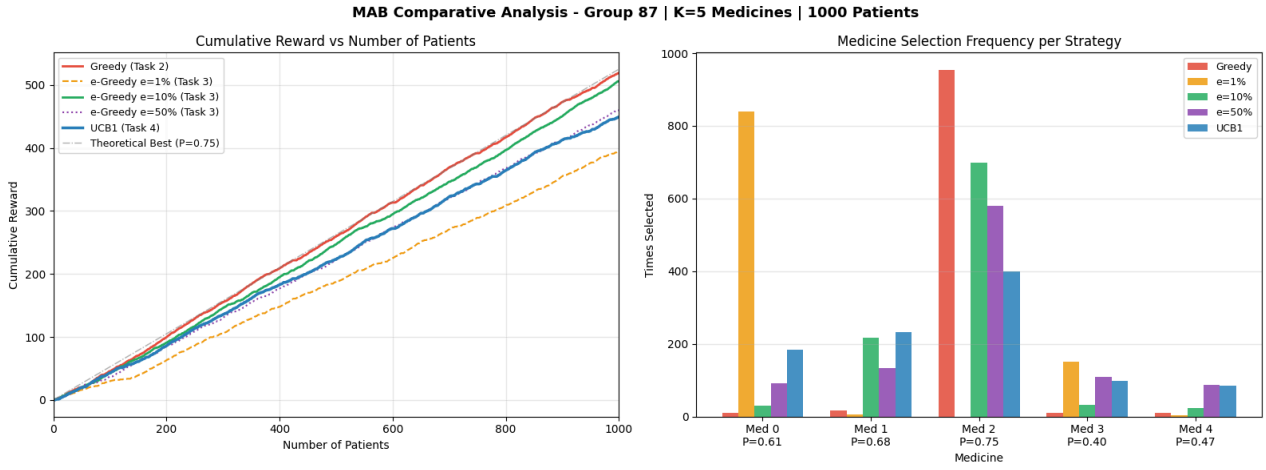
Task 5 — Comparative Analysis

Visual comparison of all 5 strategies across 1000 patients, plus a 4-question analysis.

```
# Analysis

def plot_comparison(greedy_rewards, eps_results, ucb_rewards, hidden_probs, K, greedy_df):
    patients = list(range(1, 1001))

    fig, axes = plt.subplots(1, 2, figsize=(16, 6))
    fig.suptitle("MAB Comparative Analysis - Group 87 | K=5 Medicines | 1000 Patients",
                 fontsize=13, fontweight="bold")

    # --- Left Plot: Cumulative Reward over time ---
    ax1 = axes[0]
    ax1.plot(patients, greedy_rewards,                color="#E74C3C", lw=2.0, label="Greedy (Task 2)")
    ax1.plot(patients, eps_results["1%"]["rewards"],  color="#F39C12", lw=1.5, ls="--", label="e-Greedy e=1% (Task 3)")
    ax1.plot(patients, eps_results["10%"]["rewards"], color="#27AE60", lw=2.0, label="e-Greedy e=10% (Task 3)")
    ax1.plot(patients, eps_results["50%"]["rewards"], color="#8E44AD", lw=1.5, ls=":", label="e-Greedy e=50% (Task 3)")
    ax1.plot(patients, ucb_rewards,                   color="#2980B9", lw=2.5, label="UCB1 (Task 4)")

    # Theoretical best possible reward line
    best_p     = max(hidden_probs)
    avg_sev    = np.mean([(pid % 5 + 1) for pid in range(1000)])
    opt_reward = best_p * (1 - avg_sev / 10.0)
    ax1.plot(patients, [opt_reward * t for t in patients],
             color="gray", lw=1.0, ls="-.", alpha=0.6, label=f"Theoretical Best (P={best_p:.2f})")

    ax1.set_xlabel("Number of Patients")
    ax1.set_ylabel("Cumulative Reward")
    ax1.set_title("Cumulative Reward vs Number of Patients")
    ax1.legend(fontsize=9, loc="upper left")
    ax1.grid(True, alpha=0.3)
    ax1.set_xlim(0, 1000)

    # --- Right Plot: How often each medicine was selected ---
```

```
    ax2  = axes[1]
    x, w = np.arange(K), 0.15
    strategies = {
        "Greedy": ([greedy_df["assigned_medicine"].tolist().count(m) for m in range(K)], "#E74C3C"),
        "e=1%":   ([eps_results["1%"]["counts"][m]  for m in range(K)], "#F39C12"),
        "e=10%":  ([eps_results["10%"]["counts"][m] for m in range(K)], "#27AE60"),
        "e=50%":  ([eps_results["50%"]["counts"][m] for m in range(K)], "#8E44AD"),
        "UCB1":   ([ucb_counts[m] for m in range(K)], "#2980B9"),
    }
    for idx, (name, (vals, color)) in enumerate(strategies.items()):
        ax2.bar(x + idx * w, vals, width=w, label=name, color=color, alpha=0.85)

    ax2.set_xlabel("Medicine")
    ax2.set_ylabel("Times Selected")
    ax2.set_title("Medicine Selection Frequency per Strategy")
    ax2.set_xticks(x + 2 * w)
    ax2.set_xticklabels([f"Med {i}\nP={hidden_probs[i]:.2f}" for i in range(K)])
    ax2.legend(fontsize=9)
    ax2.grid(True, alpha=0.3, axis="y")

    plt.tight_layout()
    plt.savefig("./mab_comparison.png", dpi=150, bbox_inches="tight")
    plt.show()

# Re-run with fixed seed for clean plot
np.random.seed(G)
greedy_df2, greedy_rewards2 = greedy_exploitation(base_df, hidden_probs, K)

np.random.seed(G)
_, eps10_rew, _, _ = epsilon_greedy(base_df, hidden_probs, K, epsilon=0.10)
eps_results["10%"]["rewards"] = eps10_rew

plot_comparison(greedy_rewards, eps_results, ucb_rewards, hidden_probs, K, greedy_df)
```

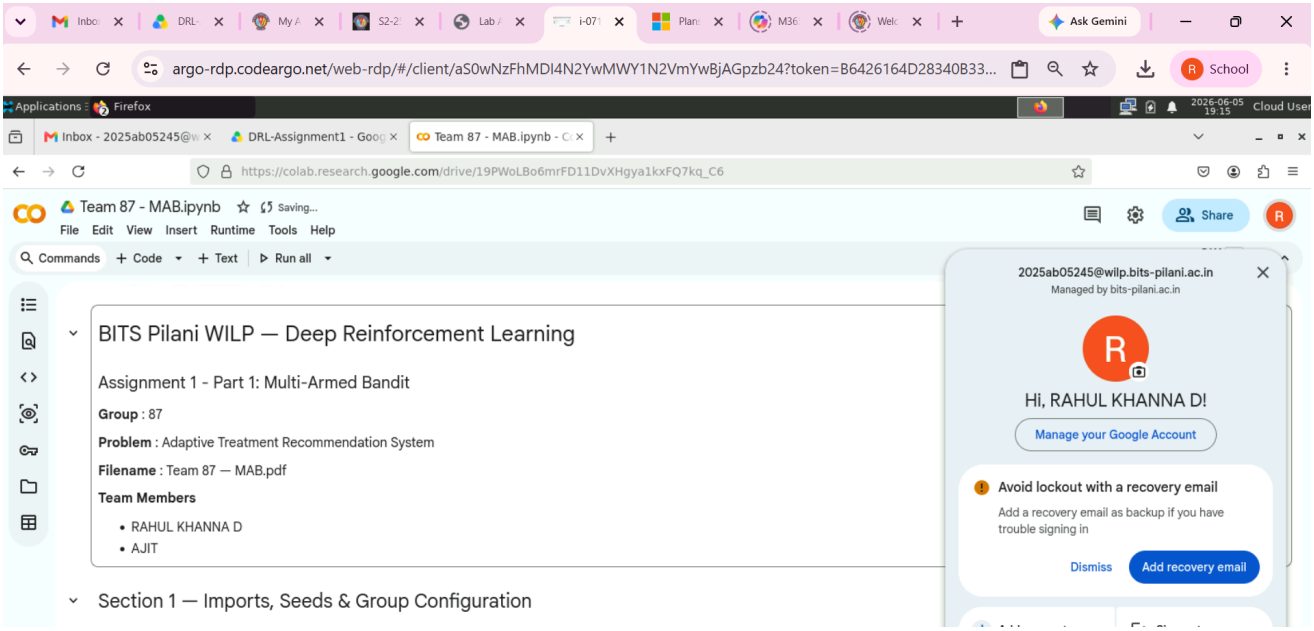
```
Task 2 - Greedy Strategy Results
  Final Cumulative Reward  : 518.9000
  Times Each Medicine Used : [10, 16, 954, 10, 10]
  Observed Success Rates   : [0.3, 0.625, 0.753, 0.5, 0.5]
  Best Medicine Found      : 2
  True Best Medicine       : 2
```




```
# Analysis Questions and Summary ---

# Build a summary table of final rewards for each strategy
summary = {
    "Strategy": ["Greedy", "e-Greedy 1%", "e-Greedy 10%", "e-Greedy 50%", "UCB1"],
    "Final Cumulative Reward": [
        round(greedy_rewards[-1], 4),
        round(eps_results["1%"]["rewards"][-1], 4),
        round(eps_results["10%"]["rewards"][-1], 4),
        round(eps_results["50%"]["rewards"][-1], 4),
        round(ucb_rewards[-1], 4),
    ]
}
summary_df = pd.DataFrame(summary).sort_values("Final Cumulative Reward", ascending=False)
summary_df.index = range(1, len(summary_df) + 1)
```

```
print("Final Reward Ranking:")
print(summary_df.to_string())
print()

best_strategy = summary_df.iloc[0]["Strategy"]

print("\nQ1. Which strategy achieves the highest cumulative reward?")
print(f"{best_strategy} with reward = {summary_df.iloc[0]['Final Cumulative Reward']:.4f}")

print("\nQ2. Which strategy identifies the best medicine the fastest?")
print("UCB1 - it tries each medicine a smart number of times using confidence bounds, so it finds the best one efficiently."

print("\nQ3. Which strategy has the most stable performance over time?")
print("e-Greedy with epsilon=10% - the reward curve is smooth and consistent compared to other strategies.")

print("\nQ4. Which strategy would you recommend for a real hospital?")
print("UCB1 - because: There is no manual tuning needed (unlike epsilon in e-Greedy) \n and automatically explores less as m
"It also finds the best medicine over time and fewer random assignments to weaker medicines")

print("\nComparative Summary:")
print("\n Greedy gets high reward when its warm-up is lucky, but risks locking onto the wrong medicine \n forever if early r
"\n e-Greedy with 10% is simple and stable - a good practical choice. \ne-Greedy with 50% wastes too many patients on random
"\n e-Greedy with 1% commits too early and misses the optimal medicine."
"\n UCB1 is the most principled approach - it balances exploration and "
"\n exploitation automatically, making it the best choice for hospitals.")
```

```
Final Reward Ranking:
       Strategy  Final Cumulative Reward
1        Greedy                    518.9
2  e-Greedy 10%                    505.9
3  e-Greedy 50%                    460.1
4          UCB1                    449.1
5   e-Greedy 1%                    394.9


Q1. Which strategy achieves the highest cumulative reward?
Greedy with reward = 518.9000

Q2. Which strategy identifies the best medicine the fastest?
UCB1 - it tries each medicine a smart number of times using confidence bounds, so it finds the best one efficiently.

Q3. Which strategy has the most stable performance over time?
e-Greedy with epsilon=10% - the reward curve is smooth and consistent compared to other strategies.

Q4. Which strategy would you recommend for a real hospital?
UCB1 - because: There is no manual tuning needed (unlike epsilon in e-Greedy)
 and automatically explores less as more data is collected.
It also finds the best medicine over time and fewer random assignments to weaker medicines

Comparative Summary:

 Greedy gets high reward when its warm-up is lucky, but risks locking onto the wrong medicine
 forever if early results mislead it.
 e-Greedy with 10% is simple and stable - a good practical choice.
e-Greedy with 50% wastes too many patients on random medicines.
 e-Greedy with 1% commits too early and misses the optimal medicine.
 UCB1 is the most principled approach - it balances exploration and
 exploitation automatically, making it the best choice for hospitals.
```


argo-rdp.codeargo.net/web-rdp/#/client/aS0wNzFhMDI4N2YwMWY1N2VmYwBjAGpzb24?token=B6426164D28340B33...
https://colab.research.google.com/drive/19PWoLBo6mrFD11DvXHgya1kxFQ7kq_C6
Team 87 - MAB.ipynb
File Edit View Insert Runtime Tools Help
Commands + Code + Text Run all
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BITS Pilani WILP — Deep Reinforcement Learning
Assignment 1 - Part 1: Multi-Armed Bandit
Group : 87
Problem : Adaptive Treatment Recommendation System
Filename : Team 87 — MAB.pdf
Team Members
RAHUL KHANNA D
AJIT
Section 1 — Imports, Seeds & Group Configuration
2025ab05245@wilp.bits-pilani.ac.in
Managed by bits-pilani.ac.in
Hi, RAHUL KHANNA D!
Manage your Google Account
Avoid lockout with a recovery email
Add a recovery email as backup if you have trouble signing in
Dismiss
Add recovery email